# Lagrangians

Nishtha Tikalal

Abstract. Quantum Field Theory (QFT) provides the essential theoretical framework that unifies quantum mechanics and special relativity. These notes explore the foundational concepts and mathematical machinery required to describe systems with infinite degrees of freedom, where particles are understood as localized excitations of underlying fields. Beginning with the transition from classical to quantum field formalisms, the text develops the canonical and path integral methods of quantization. Core topics include the analysis of free and interacting fields, the derivation of scattering amplitudes, and the implementation of diagrammatic perturbation theory. Symmetries, conservation laws, and the techniques necessary to handle regularized infinities are examined systematically. Ultimately, this material establishes the universal language used to analyze complex, multi-particle interactions across both high-energy and many-body physics.

Contents

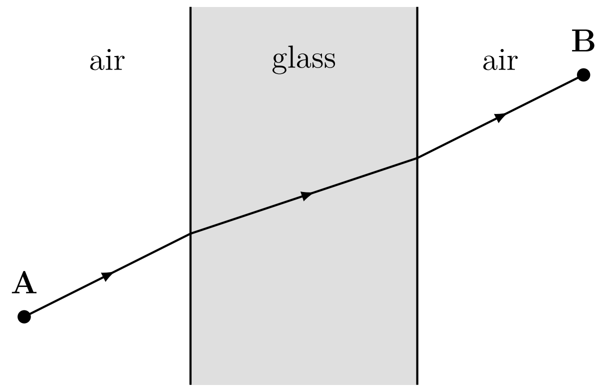


## 1. Lagrangians

### 1.1. Fermat's Principle

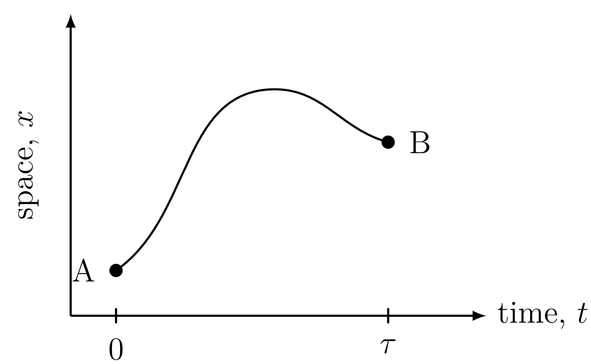
Fermat's principle stems from the study of optics. The situation is quite simple: just imagine a ray of light passing through a slab of glass. The bending of the light near the material interface can be calculated using *Snell's Law*. Fermat was able to derive Snell's Law using the **principle of least time**. This is what we previously defined as the *principle of least action*.

**Definition 1.1.** The **principle of least time** states that the path taken between two points by a ray of light is the path that is travelled in the shortest amount of time. Since light travels faster in air than any medium, anytime it crossed a medium it would do so at a steep angle to avoid lengthy travel distances.




### 1.2. Newton's Laws

Similar concepts of Fermat's principle is seen in Newtonian mechanics. We can imagine a particle of mass $m$ experiencing a force $F$ that moves solely in the x–direction from one point to another.




Using Newton's second law,

$$F = m\ddot{x}, \tag{1}$$

we can perform integration twice to yield the path of the particle $x(t)$. On thing to note immediately is how such methods disagree with *quantum mechanics*. Newtonian mechanics constructs the placement of a particle at each time. Quantum mechanics specifies that while you can find where the particle started and ended, you have no way of knowing what it did in–between.

Having some sort of "quantum–mechanical friendly" middle ground is essential, and that is what birthed the Lagrangian. Recall the equations for kinetic, potential, and total energy respectively.

$$\bar{T} = \frac{1}{\tau}\int_0^\tau \frac{1}{2}m[\dot{x}(t)]^2\,dt. \tag{2}$$

The average potential energy $\bar{V}$ during the trajectory is given by

$$\bar{V} = \frac{1}{\tau}\int_0^\tau V[x(t)]\,dt. \tag{3}$$

### 1.3. Functionals

**Definition 1.2.** A **function** is a machine that turns one number into another number.

**Definition 1.3.** A **functional** is a machine that turns a function into another number.

**Example 1.4.** An example of a functional is given by

$$F[f] = \int_0^1 f(x)\,\mathrm{d}x. \tag{4}$$

For the function $f(x) = x^2$, the functional returns:

$$F[f] = \int_0^1 x^2\,\mathrm{d}x = \frac{1}{3}. \tag{5}$$

To observe how a functional changes as the function is varied, we use **functional differentiation**. Recall that the definition (as a limit) is: The derivative of a function is defined as follows:

$$\frac{\mathrm{d}f}{\mathrm{d}x} = \lim_{\epsilon\to 0}\frac{f(x+\epsilon)-f(x)}{\epsilon}. \tag{6}$$

Functional differentiation measures the linear response of a functional to an infinitesimal localized change (a spike) in its input function at a specific point. The formal definition of a functional derivative of a functional $F[f(y)]$ with respect to the function evaluated at a target point $x$ is given by the limit:

$$\frac{\delta F[f]}{\delta f(x)} = \lim_{\epsilon\to 0}\frac{F[f(y)+\epsilon\delta(y-x)]-F[f(y)]}{\epsilon} \tag{7}$$

where $\delta(y-x)$ is the Dirac delta function centered at $x$.

**Example 1.5.** Evaluate the functional derivative of the time-averaged potential functional:

$$\bar{V}[x] = \frac{1}{\tau}\int_0^\tau V[x(t)]\,dt$$

with respect to the path coordinate $x(t)$.

*Solution Step-by-Step.* **Step 1: Clarifying the Notation**

To avoid confusing the target variable with the dummy integration variable, we rewrite the target point of differentiation as $t_0$. We want to find:

$$\frac{\delta\bar{V}[x]}{\delta x(t_0)}$$

**Step 2: Apply the Definition**

Substitute our specific functional $\bar{V}[x]$ into the general functional derivative limit formula. This requires replacing the original path $x(t)$ with a varied path $x(t)+\epsilon\delta(t-t_0)$:

$$\frac{\delta\bar{V}[x]}{\delta x(t_0)} = \lim_{\epsilon\to 0}\frac{1}{\epsilon}\left[\frac{1}{\tau}\int_0^\tau V\left(x(t)+\epsilon\delta(t-t_0)\right)dt - \frac{1}{\tau}\int_0^\tau V[x(t)]\,dt\right]$$

**Step 3: Taylor Expand the Potential Term**

Since $\epsilon\to 0$, we can linearly approximate the potential function $V$ using a first-order Taylor series expansion about the unperturbed path $x(t)$:

$$V\left(x(t)+\epsilon\delta(t-t_0)\right) \approx V[x(t)] + V'[x(t)]\cdot\epsilon\delta(t-t_0) + \mathcal{O}(\epsilon^2)$$

where $V'[x(t)] = \frac{\partial V}{\partial x}$ evaluated at the path position at time $t$.

**Step 4: Substitute and Cancel**

Insert this Taylor expansion back into the limit expression:

$$\frac{\delta \bar{V}[x]}{\delta x(t_0)} = \lim_{\epsilon \to 0} \frac{1}{\epsilon} \left[ \frac{1}{\tau} \int_0^\tau \Big( V[x(t)] + \epsilon V'[x(t)] \delta(t - t_0) \Big) \, dt - \frac{1}{\tau} \int_0^\tau V[x(t)] \, dt \right]$$

By linearity of integration, the unperturbed potential terms cancel out perfectly:

$$\frac{\delta \bar{V}[x]}{\delta x(t_0)} = \lim_{\epsilon \to 0} \frac{1}{\epsilon} \left[ \frac{\epsilon}{\tau} \int_0^\tau V'[x(t)] \delta(t - t_0) \, dt \right]$$

The scalar parameter $\epsilon$ in the numerator and denominator divides out:

$$\frac{\delta \bar{V}[x]}{\delta x(t_0)} = \frac{1}{\tau} \int_0^\tau V'[x(t)] \delta(t - t_0) \, dt$$

**Step 5: Evaluate the Dirac Delta Integral**

The sifting property of the Dirac delta function states that $\int f(t)\delta(t - t_0)dt = f(t_0)$ if $t_0$ is inside the integration bounds. Assuming $0 \le t_0 \le \tau$, the integral collapses to the value of the integrand evaluated precisely at $t = t_0$:

$$\frac{\delta \bar{V}[x]}{\delta x(t_0)} = \frac{1}{\tau} V'[x(t_0)]$$

Dropping the zero subscript to match the standard shorthand notation used in the textbook yields the final result:

$$\frac{\delta \bar{V}[x]}{\delta x(t)} = \frac{1}{\tau} V'[x(t)]$$

□

**Example 1.6.** Evaluate the functional derivative of the time-averaged kinetic energy functional:

$$\bar{T}[x] = \frac{1}{\tau} \int_0^\tau \frac{1}{2} m [\dot{x}(t)]^2 \, dt$$

with respect to the path coordinate $x(t)$, where $\dot{x}(t) = \frac{dx}{dt}$.

*Solution Step-by-Step.* **Step 1: Shift the Path and its Velocity**

To avoid notation clashing, let the target point of differentiation be $t_0$. We want to find $\frac{\delta \bar{T}[x]}{\delta x(t_0)}$.

We vary the path: $x(t) \to x(t) + \epsilon\delta(t - t_0)$. Taking the time derivative of this varied path gives the varied velocity:

$$\dot{x}(t) \to \dot{x}(t) + \epsilon \frac{d}{dt} \delta(t - t_0)$$

**Step 2: Apply the Functional Derivative Definition**

Substitute the varied velocity into the functional definition:

$$\frac{\delta \bar{T}[x]}{\delta x(t_0)} = \lim_{\epsilon \to 0} \frac{1}{\epsilon} \left[ \frac{1}{\tau} \int_0^\tau \frac{1}{2} m \left( \dot{x}(t) + \epsilon \frac{d}{dt} \delta(t - t_0) \right)^2 dt - \frac{1}{\tau} \int_0^\tau \frac{1}{2} m [\dot{x}(t)]^2 dt \right]$$

**Step 3: Expand and Isolate the Linear Term**

Expand the squared term inside the first integral:

$$\left(\dot{x}(t) + \epsilon \frac{d}{dt}\delta(t - t_0)\right)^2 = [\dot{x}(t)]^2 + 2\epsilon\dot{x}(t)\frac{d}{dt}\delta(t - t_0) + \mathcal{O}(\epsilon^2)$$

Substitute this back into the limit. The unperturbed $[\dot{x}(t)]^2$ terms cancel out, and dividing by $\epsilon$ removes the parameter from the linear term:

$$\frac{\delta\bar{T}[x]}{\delta x(t_0)} = \frac{1}{\tau}\int_0^\tau m\dot{x}(t)\frac{d}{dt}\delta(t - t_0)\,dt$$

**Step 4: Integration by Parts**

We cannot directly evaluate this integral because the derivative is acting on the delta function. We use integration by parts ($\int u\,dv = uv - \int v\,du$) to move the time derivative onto $\dot{x}(t)$:

$$u = m\dot{x}(t) \implies du = m\ddot{x}(t)\,dt$$
$$dv = \frac{d}{dt}\delta(t - t_0)\,dt \implies v = \delta(t - t_0)$$

Applying the formula:

$$\frac{\delta\bar{T}[x]}{\delta x(t_0)} = \frac{1}{\tau}\left[\Big[m\dot{x}(t)\delta(t - t_0)\Big]_0^\tau - \int_0^\tau m\ddot{x}(t)\delta(t - t_0)\,dt\right]$$

Since the variations must vanish at the boundaries ($t = 0$ and $t = \tau$), the delta function evaluates to zero at the limits, making the boundary term vanish:

$$\frac{\delta\bar{T}[x]}{\delta x(t_0)} = -\frac{1}{\tau}\int_0^\tau m\ddot{x}(t)\delta(t - t_0)\,dt$$

**Step 5: Evaluate the Delta Integral**

Using the sifting property of the Dirac delta function, the integral picks out the value of the integrand at $t = t_0$:

$$\frac{\delta\bar{T}[x]}{\delta x(t_0)} = -\frac{m\ddot{x}(t_0)}{\tau}$$

Dropping the zero subscript to yield the textbook's standard notation:

$$\frac{\delta\bar{T}[x]}{\delta x(t)} = -\frac{m\ddot{x}}{\tau}$$

$\square$

**1.4. Lagrangian and Least Action**

Recall eqs. (1.3) and (1.3). From newtonian mechanics, we know that $F = m\ddot{x}$. We also know that from electromagnetism, $F = -\frac{dV}{dx}$. Thus, $m\ddot{x} = -\frac{dV}{dx}$, and therefore for variations about a classical trajectory,

$$\frac{1}{\tau}\frac{dV}{dx} = -\frac{m\ddot{x}}{\tau} \implies m\ddot{x} = -\frac{dV}{dx} \tag{8}$$

We can re–write eq. (8) as:

$$\frac{\delta}{\delta x(t)}\left(\bar{T}[x]-\bar{V}[x]\right)=0, \tag{9}$$

which tells us that the *difference* between the average kinetic and potential energy is stationary about a classical trajectory. You might recognize, but this helps us to define the Lagrangian:

$$L=T-V \tag{10}$$

Upon integration of eq. (10), we obtain **action**:

$$S=\int_0^\tau L\,\mathrm{d}t, \tag{11}$$

Further expressing eq. (11),

$$\begin{aligned}\frac{\delta S}{\delta x(t)} &= \frac{\delta}{\delta x(t)}\left[\tau\bar{T}[x]-\tau\bar{V}[x]\right]\\ &= \tau\frac{\delta\bar{T}[x]}{\delta x(t)}-\tau\frac{\delta\bar{V}[x]}{\delta x(t)}\\ &= \tau\left(-\frac{m\ddot{x}}{\tau}\right)-\tau\left(\frac{1}{\tau}\frac{\mathrm{d}V}{\mathrm{d}x}\right)\\ &= -m\ddot{x}-\frac{\mathrm{d}V}{\mathrm{d}x}\\ &= -\left(m\ddot{x}+\frac{\mathrm{d}V}{\mathrm{d}x}\right)\\ &= 0\end{aligned}$$

Eq. (**??**) is known as **Hamilton's principle of least action**.

**Definition 1.7. Hamilton's principle of least action** states that the clasical trajectory taken by a particle is such that the action is stationary. Small adjustments will only increase the action.

The Lagrangian $L$ can also be written as a function of both position and velocity, which we know as the **Euler–Lagrange** equation.

$$\frac{\delta L}{\delta x(t)}-\frac{\mathrm{d}}{\mathrm{d}t}\frac{\delta L}{\delta\dot{x}(t)}=0, \tag{12}$$

The Lagrangian density $]mathscrL$ is related to $L$ by:

$$L=\int\mathrm{d}x\,\mathcal{L}, \tag{13}$$

and the action is given by:

$$S=\int\mathrm{d}t\,L=\int\mathrm{d}t\,\mathrm{d}x\,\mathcal{L}. \tag{14}$$

Relativistically, we can re–write eq. (14) as:

$$S=\int\mathrm{d}^4x\,\mathcal{L}(\phi,\partial_\mu\phi). \tag{15}$$

$$\boxed{\frac{\delta S}{\delta\phi}=\frac{\partial\mathcal{L}}{\partial\phi}-\partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\right)=0,} \tag{16}$$

## REFERENCES

[1] T. Lancaster and S. J. Blundell, *Quantum Field Theory for the Gifted Amateur*, Oxford University Press, 2014.